21. When Does a RAG Framework Actually Matter? A Systematic Review and Controlled Experimental Comparison Project Proposal for NLP Course, Winter 2026/27

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1 Introduction

Retrieval-augmented generation (RAG) is a technique that helps large language models to retrieve and incorporate new information from external data sources, that supplement knowledge from the LLM's pre-existing training data. Many frameworks such as LangChain, LlamaIndex, Haystack, offer seemingly similar capabilities. They have different level of abstraction, default pipelines, component choices. Benchmarks comparing some frameworks usually changes it together with embedding model, retrieved context, prompts and some other parameters. Then it is unclear what caused the performance to differ, the framework itself, or its default settings. The project tests how different retrieval and answer quality of popular RAG frameworks is, when everything except their models, context, embeddings and prompts held constant.

2 Data

2.1 Quasper

READY Question-answering systems make it easier to find answers when someone is searching for information. However, developing such tools requires appropriate test datasets that accurately reflect the complexity of the problems involved. Many existing datasets for question-based information retrieval are limited to simple, factual queries. Quasper is a dataset designed to bridge this gap, it contains 5,049 questions based on 1,585 scientific papers in the field of natural language processing. The questions and answers were prepared by individuals with expertise in NLP. First, they read the title and abstract of a paper and formulated a question concerning the work as a whole. Subsequently, a different group provided answers, complete with justifications and references to specific excerpts from the text. The results indicate that models performing well on other datasets achieve significantly lower scores on this one. Consequently, this dataset serves as an effective tool for evaluating and comparing various RAG (Retrieval-Augmented Generation) frameworks. The data includes the 'id ' and the following labels: 'title', and 'abstract' (of type string) and 'full_text', 'qas' (of type sequence). Each document contains an average of 3.2 questions. The 'qasper-dev-v0.1.json' file, which contains 281 rows, will be used as a benchmark set. It is licensed under CC BY 4.0 and was obtained from Hugging Face.

2.2 HotpotQA

READY HotpotQA is a dataset containing 113,000 question-answer pairs that require multi-hop reasoning, based on Wikipedia. It was created by an NLP research team. The dataset includes the following labels: 'answer', 'context', 'id', 'level', 'question', 'supporting_facts', and 'type'. From the "distractors" subset, we select 150 questions for the fixed test set. The HotpotQA dataset is distributed under the CC BY-SA 4.0 license, and the data was obtained from the Hugging Face platform.

3 Frameworks

3.1 Haystack

Haystack is an open-source framework for building AI agents and RAG applications (Pietsch et al., 2024). Its core characteristic is an architecture based on pipelines, which together with components (nodes) and other elements such as Agents provide basis for an observable, clean structure. The connections between components are strongly typed, which further supports that goal. The

framework also offers built-in tracing support. It also features its own RAG evaluators while simultaneously supporting adding external ones.

3.2 ARES: An Automated Evaluation Framework for RAGs

ARES is a publicly sourced framework designed for evaluating RAG performance (Saad-Falcon et al., 2024). The scoring and comparisons it makes grounds for are based on context relevance, answer faithfulness, and answer relevance. ARES uses light-weight LLM judges and prediction-powered inference (PPI). It is designed to rely on synthethic training data and only a small set of human-made annotations. In their work the original researchers have compared ARES to RAGAS and GPT-3.5 Judge. They compared these judges against selected datasets (among them Natural Questions and HotpotQA). Only a couple hundred human-made annotations needed to be used. They found ARES had achieved the most accurate evaluation results of the three RAG judges across all of the explored datasets. Other reviewers have mentioned excellent precision of the framework but also noted that the remaining reliance on human-made annotations might possibly restrict scalability (Brown et al., 2025).

3.3 RAGFlow

RAGFlow is an open-source RAG engine based on deep document understanding. When combined with an LLM, it provides answers to questions, complete with substantiated citations from various data formats. RAGFlow enables developers to transform data into high-fidelity, production-ready AI systems with higt efficiency and precision. The model analyzes the question and, if it needed, breaks it down, retrieves knowledge, and verifies evidence. It complete context through many rounds of retrieval and reasoning to help generate well-grounded answers. It choose beetween thinking modes to control retrieval and reasoning depth according to question complexity. RAGFlow supports text, presentaions, images PDFs and other data types.

3.4 DSPy: Declarative Self-improving Python

DSPy is an opensource framework for programming—rather than prompting—language models. It helps build AI systems and optimize their prompts and weights—from simple classifiers to complex RAG pipelines and agents.DSPy stands for Declarative Self-improving Python. It can be used to Python code and then teach LM to deliver high-quality outputs. It compose programs with primitives such as signatures, modules and optimizers.

4 Data statement

The QASPER and HotpotQA are open-access datasets licensed under CC BY 4.0. The datasets contain no private personal data. We process and use them in accordance with data privacy standards. The Haystack, ARES, RAGFlow, DSPy frameworks are open source and on public repositories on GitHub.

5 Solution concept and plan

5.1 Hypotheses

- **H1:** retrieval helps over a no-retrieval LLM more on Qasper (specialised, long scientific papers) than on HotpotQA (Wikipedia, likely seen in pre-training).
- **H2:** with a matched configuration (same chunker, embedder, k, prompt, generator), differences between frameworks stay within run-to-run spread, and configuration explains more variance than framework identity.
- **H3:** frameworks differ clearly in cost (latency, memory, tokens) even when quality is equal.

5.2 Pipeline

Figure 1 shows the pipeline. Every framework and an own minimal implementation run on the identical grid of chunk size, retriever, reranker and k; corpus, questions, generator and prompt are fixed. Questions come from Qasper and HotpotQA (Section 2). For each dataset, all source documents are pooled into one shared index, so that retrieval is a real search rather than a choice among a few paragraphs. Baselines: a no-retrieval LLM and a BM25-only RAG without any framework.

5.3 Metrics

- **Retrieval:** Recall@5 (primary), MRR, nDCG@10.
- **Answer:** exact match and token F1 (primary).

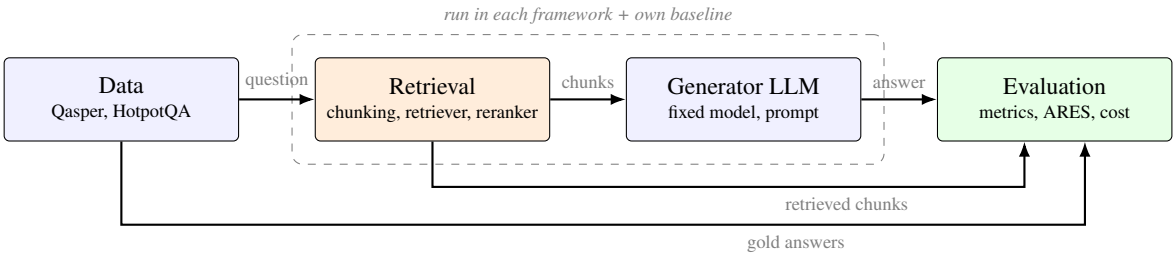


Figure 1: Experimental pipeline. The orange stage is varied in a grid; blue stages are fixed. Each framework is run on the identical grid with the same data and metrics. Retrieval metrics use the retrieved chunks, answer metrics, ARES and cost use the generated answers. Baselines (no retrieval, BM25 only) are described in the text.

- **Faithfulness:** ARES score (Saad-Falcon et al., 2024), the same evaluator for all frameworks (not Haystack's built-in ones), checked against about 150 items labelled by the team (Cohen's κ) and used only if $\kappa \geq 0.6$.
- **Cost:** latency, peak memory and tokens per query.
- **Aggregation:** macro-average per dataset and split; failures score 0.
- **Uncertainty:** mean $\pm$ std over at least 3 seeds with 95% bootstrap confidence intervals; a difference between frameworks is claimed only if the interval excludes 0.

5.4 Success deliverables

- **D1:** one reproducible repository running at least 2 frameworks and 2 baselines on at least 2 datasets over the full grid with at least 3 seeds, from which a stranger reproduces one result using the README alone.
- **D2:** a quantified answer to H1–H3 (effect sizes with intervals), a decision rule ("use a framework / use plain RAG / skip retrieval") and an error analysis of at least 20 failures grouped into named types.

5.5 Work plan and owners

Jan owns the pipeline core, the grid runner, compute and the Haystack adapter; Karolina the data, the RAGFlow adapter, literature and error analysis; Anna the metrics, ARES, statistics and the report. Each result is reviewed by the next person in the cycle Jan $\rightarrow$ Karolina $\rightarrow$ Anna $\rightarrow$ Jan. Table 1 gives the week-by-week plan.

6 Contribution table

Student	Work	Hours
Jan Czemko	Solution concept and plan	3h
Karolina Móżdżyńska	cell5	cell6
Anna Piórkowska	Introduction, Data, RAGFLow and DSPy	4h

7 Declaration of LLM use

AI was used for creating LaTeX tables and figures, as well as general spell checking and grammar correction/improvment. It was used to search for literature and sources of knowledge.

References

Pradeep Dasigi, Kyle Lo, Iz Beltagy, Arman Cohan, Noah A. Smith, and Matt Gardner. 2021. A Dataset for Question Answering on Scientific Papers. In *NAACL-HLT 2021*.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. In *EMNLP 2018*.

Andrew Brown, Muhammad Roman, Barry Devereux. 2025. A Systematic Literature Review of Retrieval-Augmented Generation: Techniques, Metrics, and Challenges.

Jon Saad-Falcon, Omar Khattab, Christopher Potts, Matei Zaharia. 2025. ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems.

Pietsch, M., Soni, T., Chan, B., Möller, T., Kostić, B. 2020. Haystack. GitHub. https://github.com/deepset-ai/haystack

Week	Jan	Karolina	Anna
W1: 5–11 Nov	repository, config schema, Haystack pipeline end to end on 5 questions	Qasper and HotpotQA loaders, fixed seeded question lists	metric code (token F1, exact match, Recall@5, MRR, nDCG@10), unit tests on toy data
W2: 12–18 Nov *18 Nov: mid-term video*	BM25-only baseline; proof of concept: Haystack and BM25 on 50 Qasper questions with token F1 and latency	Qasper evidence-to-chunk mapping, EDA start, literature table	video script and recording; check on a sample of 50 Qasper questions that the shared index retrieves the right paper
W3: 19–25 Nov *25 Nov: Q&A*	no-retrieval baseline; **budget estimate** (calls, hours, cost) for the full grid	RAGFlow adapter (pair session with Jan)	ARES setup and judge training; rebuttal; draft analysis plan (tests, variance method for H2)
W4: 26 Nov–2 Dec *2 Dec: report*	pilot grid (10% of questions, 1 seed) for both frameworks; runner resumes after failure; README draft	parity check of both frameworks and the own baseline under the matched configuration (H2); EDA report; revised literature	**freeze analysis plan** (tag in repo: metrics, tests, variance method); report
W5: 3–9 Dec *9 Dec: feedback meeting*	launch the full grid, seed 1, Qasper	sample 150 items for ARES labelling (stratified over frameworks, disjoint from calibration); label	label the same items independently; ARES calibration on the disjoint set
W6: 10–16 Dec	full grid, seed 1, HotpotQA; first sanity table	error analysis on seed-1 outputs, draft of failure types	κ computed; **gate 16 Dec**: ARES used only if $\kappa \geq 0.6$; dry run of the statistics on seed 1
W7: 17–23 Dec	seeds 2–3 launched; cost measurements on an idle machine	at least 20 failures grouped into named types	statistics on seed 1; draft tables and figures for H1–H3
W8: 24–30 Dec	buffer: runs continue unattended, Jan monitors	light drafting, no deadline: related-work text from the literature table	backup for the runs
W9: 31 Dec–6 Jan *6 Jan: results freeze*	seeds finished, reruns of failures, tag the frozen results	error analysis finalised on the frozen results	final statistics: effect sizes, intervals, decision rule
W10: 7–13 Jan *13 Jan: report, video, Q&A*	final README; stranger test (a classmate reproduces one result); demonstration notebook	final literature, quiz, check of the data statement, error-analysis section	report, slides, video, LLM declaration, contribution table
W11: 14–20 Jan *20 Jan: rebuttals*	own answers to final Q&A	own answers to final Q&A	own answers to final Q&A

Table 1: Weekly plan with named owners, aligned with the course deadlines (in italics). D1 and D2 are due on 13 Jan. Everyone logs hours weekly for the contribution table.

InfiniFlow. 2024. GitHub. https://github.com/infiniflow/ragflow

Omar Khattab, Stanford NLP. DSPy: The Framework for Programming—Not Prompting—Language Models. GitHub. https://github.com/stanfordnlp/dspy.

Omar Khattab, Stanford NLP. DSPy Documentation: Program, Don't Prompt, Your LLMs. https://dspy.ai/current/.